# Confluent Cloud Terraform Deployment — Timer-Driven PTF UDFs (early access example)

> This example deploys the **Session Timeout Detector** and **Per-Event Follow-Up** PTF UDFs (source in examples/ptf_udf_timer_driven/java/) to **Confluent Cloud** using Terraform. Both UDFs are packaged in a single JAR. All infrastructure (environment, Kafka cluster, Flink compute pool, service accounts, API keys) and Flink SQL statements are declared as Terraform resources.

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## 1.0 Overview

### 1.1 How this differs from the other deployment paths

| Aspect | Confluent Cloud Terraform (this example) | Confluent Platform SQL Client |
|---|---|---|
| Where it runs | Confluent Cloud | Confluent Platform on Minikube |
| How SQL is submitted | `confluent_flink_statement` Terraform resources | `sql-client.sh -f` on the JobManager pod |
| UDF JAR delivery | `confluent_flink_artifact` (uploaded to CC) | `kubectl exec` to Flink pods |
| Entry point | `make deploy-cc-ptf-udf-timer-driven` | `make deploy-cp-ptf-udf-timer-driven` |
| Requires code compilation | ✅ (Java + Gradle for UDF JAR) | ✅ (Java + Gradle for UDF JAR) |
| Statement lifecycle | Managed by Terraform state | Managed by Flink session cluster |

| Aspect | Confluent Cloud Terraform (this example) | Confluent Platform SQL Client |
|---|---|---|
| Same codebase for both? | ✅ (**same Java UDF code**, different Terraform vs SQL Client for deployment) | ✅ (**same Java UDF code**, different Terraform vs SQL Client for deployment) |

# 2.0 How it works

## 2.1 Infrastructure provisioning

Terraform declares the full Confluent Cloud stack:

| Resource | Purpose |
|---|---|
| `confluent_environment` | Isolated CC environment (`ptf-udf-timer-driven`) |
| `confluent_kafka_cluster` | Standard single-zone Kafka cluster (AWS us-east-1) |
| `confluent_schema_registry_cluster` | Stream Governance Essentials (auto-provisioned with environment) |
| `confluent_service_account` | `ptf-udf-timer-driven` service account with FlinkDeveloper, ResourceOwner, and Assigner roles |
| `confluent_flink_compute_pool` | Flink compute pool (max 10 CFU) |
| `confluent_api_key` | Flink-specific API key for statement submission |
| `flink_api_key_rotation` module | Rotating API key pair for the Flink service account |

## 2.2 JAR delivery

On Confluent Cloud, UDF JARs are uploaded as **Flink artifacts** via the `confluent_flink_artifact` resource. The JAR is built from examples/ptf_udf_timer_driven/java/ and uploaded directly from the local build output. Both `CREATE FUNCTION ... USING JAR` statements reference the same artifact using a `confluent-artifact://` URI, registering `session_timeout_detector` and `per_event_follow_up` from the single JAR.

## 2.3 Statement flow

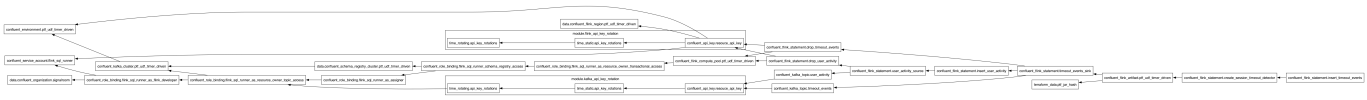
Terraform manages the SQL statements as `confluent_flink_statement` resources with explicit `depends_on` ordering:

```
┌──────────────────────────────────────────────────────────────────────┐
│  ── Named timer pipeline (SessionTimeoutDetector) ──                 │
│  Step 1:  DROP TABLE IF EXISTS user_activity            → OK         │
│  Step 2:  CREATE TABLE user_activity (...)              → OK         │
│  Step 3:  INSERT INTO user_activity VALUES (sample data) → submitted │
│  Step 4:  DROP TABLE IF EXISTS timeout_events           → OK         │
│  Step 5:  CREATE TABLE timeout_events (...)             → OK         │
```

```
│                                                                    │
│  ── Unnamed timer pipeline (PerEventFollowUp) ──                   │
│  Step 6:  DROP TABLE IF EXISTS user_actions             → OK       │
│  Step 7:  CREATE TABLE user_actions (...)               → OK       │
│  Step 8:  INSERT INTO user_actions VALUES (sample data) → submitted │
│  Step 9:  DROP TABLE IF EXISTS follow_up_events         → OK       │
│  Step 10: CREATE TABLE follow_up_events (...)           → OK       │
│                                                                    │
│  ── Shared JAR upload and function registration ──                 │
│  Step 11: Upload UDF JAR as confluent_flink_artifact    → OK       │
│  Step 12: CREATE FUNCTION session_timeout_detector      → OK       │
│  Step 13: INSERT INTO timeout_events                    → submitted │
│  Step 14: CREATE FUNCTION per_event_follow_up           → OK       │
│  Step 15: INSERT INTO follow_up_events                  → submitted │
└────────────────────────────────────────────────────────────────────┘
```

Steps 13 and 15 are **long-running streaming jobs**. They run continuously, reading from their respective source topics and writing output to sink topics.

Once deployment completes, Terraform generates a visual **resource graph** at `examples/ptf_udf_timer_driven/cc_deploy/terraform.png`, providing an at-a-glance view of the infrastructure and resource dependencies:



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## 3.0 Prerequisites

- macOS with Homebrew or Linux with apt-get
- Java 17+
- Terraform installed
- A Confluent Cloud account with a **Cloud API key** and **secret** (create one here)
- The UDF JAR must be built before deploying:

```
make build-ptf-udf-timer-driven     # builds the UDF fat JAR from
examples/ptf_udf_timer_driven/java/
```

---

## 4.0 How to run

All commands are run from the **project root** (where the `Makefile` lives).

### 4.1 Deploy

A single target builds the UDF JAR and runs `terraform apply`:

```
make deploy-cc-ptf-udf-timer-driven CONFLUENT_API_KEY=<your-key>
CONFLUENT_API_SECRET=<your-secret>
```

Behind the scenes this runs:

| Step | What it does |
|---|---|
| 1 | `./gradlew clean shadowJar` — builds the UDF fat JAR from `examples/ptf_udf_timer_driven/java/` |
| 2 | `terraform init` — initializes the Terraform working directory |
| 3 | `terraform apply -auto-approve` — provisions all CC infrastructure and submits Flink SQL statements |
| 4 | Generates a Terraform visualization at `examples/ptf_udf_timer_driven/cc_deploy/terraform.png` |

### 4.2 Monitor

Monitor the running Flink statements in the Confluent Cloud Console:

1. Navigate to your **ptf-udf-timer-driven** environment
2. Open the **Flink** tab to see compute pools and running statements
3. Open the **Topics** tab to inspect `user_activity`, `timeout_events`, `user_actions`, and `follow_up_events`

### 4.3 Tear down

To destroy all Confluent Cloud resources created by Terraform:

```
make teardown-cc-ptf-udf-timer-driven CONFLUENT_API_KEY=<your-key>
CONFLUENT_API_SECRET=<your-secret>
```

This runs `terraform destroy -auto-approve`, removing all Flink statements, the compute pool, Kafka cluster, service accounts, and the environment.

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## 5.0 Troubleshooting

If you encounter an error during `make deploy-cc-ptf-udf-timer-driven` that looks like the following, it is because your organization does not have early access to timer services for PTF UDFs. Please contact your Confluent Account Executive to request access:

```
│ Error: error waiting for Flink Statement "tf-2026-04-02-100522-54a7b07c-
4a42-4d53-90e0-034cb2ae1ea4" to provision: Flink Statement "tf-2026-04-02-
100522-54a7b07c-4a42-4d53-90e0-034cb2ae1ea4" provisioning status is
"FAILED": Method 'onTimer' of function class 'ptf.SessionTimeoutDetector'
```

```
- Timer service is not supported for PTFs.; could not parse error details;
raw response body: "{\"api_version\":\"sql/v1\",\"environment_id\":\"env-
m60dnq\",\"kind\":\"Statement\",\"metadata\":{\"created_at\":\"2026-04-
02T14:05:22.440172Z\",\"labels\":
{},\"resource_version\":\"5\",\"self\":\"https://flink.us-east-
1.aws.confluent.cloud/sql/v1/organizations/bd545cc3-8e7c-4387-b6d8-
b6d1497a9df7/environments/env-m60dnq/statements/tf-2026-04-02-100522-
54a7b07c-4a42-4d53-90e0-034cb2ae1ea4\",\"uid\":\"eff86f2f-bb53-474d-87bd-
3300ead6db9f\",\"updated_at\":\"2026-04-
02T14:05:22.64651Z\"},\"name\":\"tf-2026-04-02-100522-54a7b07c-4a42-4d53-
90e0-034cb2ae1ea4\",\"organization_id\":\"bd545cc3-8e7c-4387-b6d8-
b6d1497a9df7\",\"spec\":{\"compute_pool_id\":\"lfcp-
nnpz1v\",\"execution_mode\":\"STREAMING\",\"principal\":\"sa-
nvj798k\",\"properties\":{\"sql.current-catalog\":\"ptf-udf-timer-
driven\",\"sql.current-database\":\"ptf-udf-timer-
driven\"},\"statement\":\"CREATE FUNCTION IF NOT EXISTS
session_timeout_detector\\n  AS 'ptf.SessionTimeoutDetector'\\n  USING JAR
'confluent-artifact://cfa-q57pk2';\\n\",\"stopped\":false},\"status\":
{\"detail\":\"\",\"network_kind\":\"PUBLIC\",\"phase\":\"PENDING\",\"trait
s\":{\"connection_refs\":
[],\"is_append_only\":true,\"is_bounded\":true,\"schema\":
{},\"sql_kind\":\"CREATE_FUNCTION\",\"upsert_columns\":null},\"warnings\":
[]}}"
|
|   with confluent_flink_statement.create_session_timeout_detector,
|   on setup-confluent-flink.tf line 340, in resource
"confluent_flink_statement" "create_session_timeout_detector":
|  340: resource "confluent_flink_statement"
"create_session_timeout_detector" {
```

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## 6.0 Resources

- Confluent Terraform Provider
- confluent_flink_statement Resource
- confluent_flink_artifact Resource
- Process Table Functions (PTFs)
- Create a User-Defined Function with Confluent Cloud for Apache Flink